

UNIVERSITÀ DEGLI STUDI DI PADOVA

Department of Civil, Environmental and Architectural Engineering

Master's Degree in Mathematical Engineering
Mathematical Modelling for Engineering and Science



MASTER'S THESIS

Investigations on some models in gradient elastoplasticity

Supervisor:

Prof. Filip Rindler
University of Warwick

Candidate:

Bernardo Ardini
2146900

Co-supervisor:

Prof. Giulio Giusteri
Università di Padova

Academic Year 2025/2026

Aims and results of the thesis

Description of a mechanical system

The systems are the elements z of a separable and reflexive Banach space Z .

Generalized forces are the elements $\mathcal{F}$ of the dual space Z^* .

Equilibrium

Equilibrium is the balance

$$-\mathcal{J}(t) + \mathcal{X}(t) = 0 \quad \text{for a.e. } t \in (0, T)$$

between internal forces $\mathcal{J}: (0, T) \rightarrow Z^*$ and external $\mathcal{X}: (0, T) \rightarrow Z^*$ forces.

PROBLEM. Given $z_0 \in Z$ and $\mathcal{X}$ find z such that $z(0) = z_0$ and equilibrium holds.

Constitutive relations

Let $\Psi: Z \rightarrow \mathbb{R} \cup \{0\}$ be the free energy and $\Phi: Z \rightarrow [0, \infty]$ the dissipation potential.

Internal forces are related to the evolution $z: [0, T] \rightarrow Z$ with a constitutive relation

$$\mathcal{J}(t) \in D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \quad \text{for a.e. } t \in (0, T).$$

DEFINITION. $\xi \in \partial\Phi(v)$ if $\Phi(w) \geq \Phi(v) + \langle \xi, w - v \rangle$ for all $w \in Z$.

PROPOSITION. If Φ is convex and $\Phi(0) = 0$ then $\langle \xi, v \rangle \geq 0$ for all $\xi \in \partial\Phi(v)$.

COROLLARY. $\frac{d}{dt}\Psi(z(t)) \leq \langle \mathcal{J}(t), \dot{z}(t) \rangle = \langle \mathcal{X}(t), \dot{z}(t) \rangle.$

Doubly nonlinear equation

$$\implies D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \ni \mathcal{X}(t)$$

for a.e. $t \in (0, T)$

Some introductory notions

Developing mechanical models through the principle of virtual powers

Description of a mechanical system

The systems are the elements z of a separable and reflexive Banach space Z .

Generalized forces are the elements $\mathcal{F}$ of the dual space Z^* .

Equilibrium

Equilibrium is the balance

$$-\mathcal{J}(t) + \mathcal{X}(t) = 0 \quad \text{for a.e. } t \in (0, T)$$

between internal forces $\mathcal{J}: (0, T) \rightarrow Z^*$ and external $\mathcal{X}: (0, T) \rightarrow Z^*$ forces.

PROBLEM. Given $z_0 \in Z$ and $\mathcal{X}$ find z such that $z(0) = z_0$ and equilibrium holds.

Constitutive relations

Let $\Psi: Z \rightarrow \mathbb{R} \cup \{0\}$ be the free energy and $\Phi: Z \rightarrow [0, \infty]$ the dissipation potential.

Internal forces are related to the evolution $z: [0, T] \rightarrow Z$ with a constitutive relation

$$\mathcal{J}(t) \in D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \quad \text{for a.e. } t \in (0, T).$$

DEFINITION. $\xi \in \partial\Phi(v)$ if $\Phi(w) \geq \Phi(v) + \langle \xi, w - v \rangle$ for all $w \in Z$.

PROPOSITION. If Φ is convex and $\Phi(0) = 0$ then $\langle \xi, v \rangle \geq 0$ for all $\xi \in \partial\Phi(v)$.

COROLLARY. $\frac{d}{dt}\Psi(z(t)) \leq \langle \mathcal{J}(t), \dot{z}(t) \rangle = \langle \mathcal{X}(t), \dot{z}(t) \rangle.$

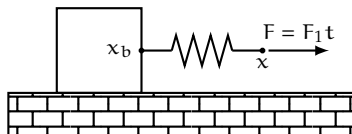
Doubly nonlinear equation

$$\implies D\Psi(z(t)) + \partial\Phi(\dot{z}(t)) \ni \mathcal{X}(t)$$

for a.e. $t \in (0, T)$

A simple example

A block on the plane with friction pulled with a spring.



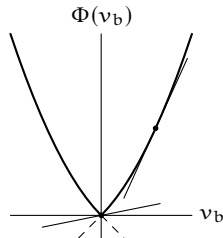
The space of configurations is $Z = \{z = (x, x_b) \in \mathbb{R}^2\}$. Let $w = (v, v_b)$ the generic rate of the system.

The free energy is $\Psi(z) = \frac{\kappa}{2}|x - x_b|^2$ while the dissipation potential is $\Phi(w) = Y|v_b| + \frac{\eta}{2}|v_b|^2$.

Start with $z(0) = (0, 0)$.

PROPOSITION. It holds that

$$\partial\Phi(v_b) = \begin{cases} [-Y, Y] & \text{if } v_b = 0 \\ \{Y \operatorname{sign} v_b + \eta v_b\} & \text{if } v_b \neq 0. \end{cases}$$



SOLUTION. It holds that $x = x_b + \frac{F}{\kappa}$. Moreover $\dot{x}_b = 0$ when $F < Y$ while $F = Y + \eta \dot{x}_b$ when $F \geq Y$.

Necking of a metallic cylindrical bar

Modeling dilatant and pressure sensitive elastoplastic materials

Classical approach

Kinematics

The space of configurations is $Z = \{(u, E_p)\}$ with the displacement $u \in H^1(\Omega, \mathbb{R}^d)$ and the plastic strain $E_p \in H^1(\Omega, \text{Sym})$. The elastic strain is $E_e = Eu - E_p$.

The generic rate is $w = (v, V_p)$ and the elastic rate is $V_e = Eu - V_p$.

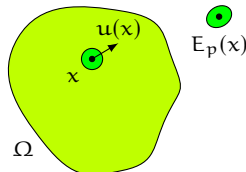
Statics

The internal generalized force is

$$\langle \mathcal{I}, w \rangle = \int_{\Omega} \left[T_e : V_e + T_p : \text{grad } V_p + K_p : \text{grad } V_p \right] dx$$

with $T_e : \Omega \rightarrow \text{Sym}$ the elastic stress, $T_p : \Omega \rightarrow \text{Sym}$ the plastic force and $K_p : \Omega \rightarrow \text{Sym} \otimes \mathbb{R}$ the plastic stress. The external generalized force is

$$\langle \mathcal{X}, w \rangle = \int_{\Omega} f \cdot v dx + \int_{\partial\Omega} h \cdot v dx.$$



PROPOSITION. The strong form of the equilibrium equation is

$$\begin{aligned} \text{div } T_e + f &= 0 \\ T_e - T_e + \text{div } K_p &= 0 \end{aligned} \quad \text{in } \Omega$$

and $T_e n = h$, $K_p n = 0$ in $\partial\Omega$.

Classical approach

Constitutive relations

Let the free energy

$$\Psi(z) = \int_{\Omega} \left[\frac{1}{2} \mathfrak{C} E_e : E_e + \frac{1}{2} \mathfrak{b} |K_p|^2 \right] dx$$

that implies

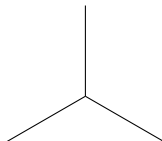
$$T_e = \mathfrak{C} E_e \quad \text{and} \quad K_p = \mathfrak{b} \operatorname{grad} E_p.$$

Yield condition and flow rule

Fix $f, g : \operatorname{Sym} \rightarrow \mathbb{R}$ convex, the yield function and plastic potential respectively. Let $C = \{f \leq 0\}$ and $L(T_p) = \{g \leq g(T_p)\}$.

It is required that $T_p \in C$ with $C \subset \operatorname{Sym}$ closed, convex and with $0 \in \operatorname{int} C$. The yield condition is $T_p \in \partial C$ and the flow rule is $\dot{E}_p = 0$ if $T_p \in \operatorname{int} C$

$$\dot{E}_p \in \begin{cases} \{0\} & \text{if } T_p \in \operatorname{int} C \\ N_{L(T_p)}(T_p) & \text{if } T_p \in \partial C. \end{cases}$$



Internal virtual power

Let τ macroscopic Kirchhoff stress and τ_p microstress

$$\tilde{W} = \int_E (\tau \tilde{\epsilon}_e + \tau_p \tilde{\epsilon}_p) V de,$$

Lemma

We can write $\tilde{W}$ as

$$\tilde{W} = - \sum_{a \in A} f_a \cdot \tilde{x}_a + \int_E (\tau_p - \tau) \tilde{\epsilon}_p V de$$

with

$$f_a = \int_E s_a q n de \quad s_a(e) = \begin{cases} 1 & \text{if } a = e_1 \\ -1 & \text{if } a = e_2 \\ 0 & \text{otherwise} \end{cases}$$

Proof.

$$\tau \tilde{\epsilon} V = \frac{v}{V} \sigma \frac{\tilde{l}}{l} V = a \sigma n \cdot (\tilde{x}_b - \tilde{x}_a)$$

□

Dirichlet, Neumann and contact conditions

We split $A = A_D \cup A_N \cup A_C$

Dirichlet condition

If $a \in A_D$ we impose $x_a = \bar{x}_a$ and call $r_a \in \mathbb{R}^d$ the Lagrange multiplier

Neumann condition

On $a \in A_N$ acts the external load $h_a \in \mathbb{R}^d$

Contact condition

If $a \in A_C$ let $\gamma_a = \phi(x_a, t)$ with $\omega^M = \{x \in \mathbb{R}^d \mid \phi(x, t) \leq 0\}$ the obstacle and $\kappa_a \in \mathbb{R}$ the Lagrange multiplier

$$\gamma_a \geq 0 \quad \kappa_a \geq 0 \quad \gamma_a \kappa_a = 0$$

External virtual power and balances

$$\tilde{\mathcal{P}} = \sum_{a \in A_D} r_a \tilde{v}_a + \sum_{a \in A_N} h_a \tilde{v}_a + \sum_{a \in A_C} \kappa_a \tilde{\gamma}_a.$$

notice that

$$\tilde{\gamma}_a = \nabla \phi(x_a, t) \tilde{v}_a$$

recall

$$\tilde{\mathcal{W}} = - \sum_{a \in A} f_a \cdot \tilde{x}_a + \int_E (\tau_p - \tau) \tilde{\epsilon}_p V de$$

Definition (Virtual Powers Principle)

equilibrium holds when $\tilde{\mathcal{W}} = \tilde{\mathcal{P}}$ for all virtual rates

Equilibrium equations

in A_D we have $f_a + r_a = 0$

in A_N we have $f_a + h_a = 0$

in A_C we have $f_a + \kappa_a \nabla \phi(x_a, t) = 0$ on

each truss $\tau_p - \tau = 0$

Elastic response

let E Young modulus

$$\tau = E \epsilon_e$$

Yield function

$$Y(\alpha) = Y_0 + H(1 - e^{-b\alpha})$$

Rate sensitivity function

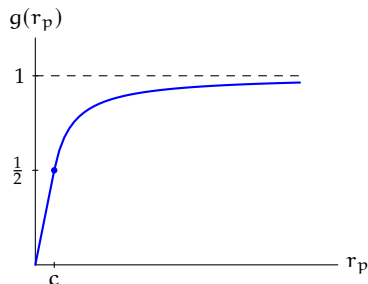
$$g(r_p) = \begin{cases} \frac{r_p}{2c} & \text{if } r_p \leq c \\ 1 - \frac{c}{2r_p} & \text{if } r_p > c, \end{cases}$$

Plastic response

$$\text{let } r_p = |\dot{\epsilon}_p|$$

$$\tau_p = g(r_p) Y(\alpha) \frac{\dot{\epsilon}_p}{r_p}$$

$$\text{with } \dot{\alpha} = |\dot{\epsilon}_p|$$



Initial boundary value problem

Problem

Let $\bar{x}_a(t)$, $h_a(t)$ and $\phi(x, t)$ be given. Find $x_a(t)$ with $x_a(0) = X_a$, $r(t)$, $\kappa(t)$, $e_p(e, t)$ with $e_p(e, 0) = 0$ and $\alpha(e, t)$ with $\alpha(e, 0) = 0$ such that for all $t \in [0, T]$

$$x_a(t) = \bar{x}_a(t) \quad \text{and} \quad f_a + r_a = 0 \quad \text{for } a \in A_D$$

$$f_a + h_a = 0 \quad \text{for } a \in A_N$$

$$f_a + \kappa_a \nabla \phi(x_a, t) = f_a + c_a = 0 \quad \text{for } a \in A_C$$

$$\gamma_a \geq 0, \quad \kappa_a \geq 0 \quad \text{and} \quad \gamma_a \kappa_a = 0 \quad \text{for } a \in A_C$$

$$\tau_p = \tau \quad \text{in } E$$

$$\dot{\alpha} = r_p \quad \text{in } E.$$

Problem

Let $\bar{\mathbf{x}}(t)$, $\mathbf{h}(t)$ and $\phi(\mathbf{x}, t)$ be given. Find $\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{X}$, $\mathbf{r}(t)$, $\boldsymbol{\kappa}(t)$, $\boldsymbol{\epsilon}_p(t)$ with $\boldsymbol{\epsilon}_p(0) = 0$ and $\boldsymbol{\alpha}(t)$ with $\boldsymbol{\alpha}(0) = 0$ such that for all t

$$\mathbf{f} + \mathbf{H}_D \mathbf{r} + \mathbf{H}_N \mathbf{h} + \mathbf{H}_C \mathbf{c} = 0$$

$$\mathbf{H}_D^T \mathbf{x} = \bar{\mathbf{x}}$$

$$\gamma_a \geq 0, \quad \kappa_a \geq 0 \quad \text{and} \quad \gamma_a \kappa_a = 0 \quad (\text{KT})$$

$$\boldsymbol{\tau}_p = \boldsymbol{\tau}$$

$$\dot{\boldsymbol{\alpha}} = \mathbf{r}_p.$$

Fischer–Burmeister function

we write the condition (KT) as

$$\boldsymbol{\Phi}(\boldsymbol{\gamma}, \boldsymbol{\kappa}) = 0 \text{ where}$$

$$\boldsymbol{\Phi}_a(\boldsymbol{\gamma}, \boldsymbol{\kappa}) = \gamma_a + c \kappa_a - |\gamma_a - c \kappa_a|$$

Residual

we write the five sets of equalities as

$\mathbf{R} = 0$ where $\mathbf{R}$ depends on $\mathbf{x}$, $\mathbf{r}$, $\boldsymbol{\kappa}$, $\boldsymbol{\epsilon}_p$, $\boldsymbol{\eta}_p$, $\boldsymbol{\alpha}$, $\dot{\boldsymbol{\alpha}}$ and t

Time discretization

At time $t - \Delta t$ we know the solution and we look for the solution at time t .
We set

$$\begin{aligned}\mathbf{x} &= \mathbf{x}^- + \Delta t \mathbf{v} \\ \epsilon_p &= \epsilon_p^- + \Delta t \eta_p \\ \alpha &= \alpha^- + \Delta t \dot{\alpha}.\end{aligned}$$

Now $\mathbf{R}$ depends on $\mathbf{v}$, $\mathbf{r}$, κ , η_p and $\dot{\alpha}$ that we collect in ξ .

Time increment

Choose Δt_0 , $\Delta t_{\max}$, $\Delta t_{\min}$, $k_{\max}$ and $k_{\min}$

$$\Delta t_k = \begin{cases} \Delta t_0 \left[\frac{\Delta t_{\max}}{\Delta t_0} \right]^{\frac{k}{k_{\max}}} & \text{if } k \geq 0 \\ \Delta t_0 \left[\frac{\Delta t_0}{\Delta t_{\min}} \right]^{\frac{k}{k_{\min}}} & \text{if } k \leq 0 \end{cases}$$

Start with $k = 0$ and decrement it if the step fails

Newton method

We solve

$$\mathbf{R}(\xi) = 0$$

with Newton method

- set $\xi_0 = \xi^-$
- impose $\mathbf{R}(\xi_{i-1} + \delta_i) \approx \mathbf{R}(\xi_{i-1}) + \mathbf{K}(\xi_{i-1})\delta_i = 0$
- solve $\delta_i = -\mathbf{K}(\xi_{i-1})^{-1}\mathbf{R}(\xi_{i-1})$
- update $\xi_i = \xi_{i-1} + \eta_i \delta_i$

Failure criteria

ITER > MAXITER

$$|\mathbf{x} - \mathbf{x}^*| > \text{TOL}(\mathbf{x})$$

$$|\mathbf{r} - \mathbf{r}^-| > \text{TOL}(\mathbf{r})$$

$$|\kappa - \kappa^-| > \text{TOL}(\kappa)$$

$$|\epsilon_p - \epsilon_p^*| > \text{TOL}(\epsilon_p)$$

$$|\alpha - \alpha^*| > \text{TOL}(\alpha).$$

with $\mathbf{x}^* = \mathbf{x}^- + \Delta t \mathbf{v}^-$ etc.

Stiffness matrix

$$\mathbf{K} = \begin{pmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} & \mathbf{K}_{13} & \mathbf{K}_{14} & 0 \\ \mathbf{K}_{21} & 0 & 0 & 0 & 0 \\ \mathbf{K}_{31} & 0 & \mathbf{K}_{32} & 0 & 0 \\ \mathbf{K}_{41} & 0 & 0 & \mathbf{K}_{44} & \mathbf{K}_{45} \\ 0 & 0 & 0 & \mathbf{K}_{54} & \mathbf{K}_{55} \end{pmatrix} = \begin{pmatrix} \frac{\partial \mathbf{f}}{\partial \mathbf{v}} + \mathbf{H}_C \frac{\partial \mathbf{c}}{\partial \mathbf{v}} & \mathbf{H}_D & \mathbf{H}_C \frac{\partial \mathbf{c}}{\partial \boldsymbol{\kappa}} & \frac{\partial \mathbf{f}}{\partial \boldsymbol{\eta}_p} & 0 \\ \mathbf{H}_D^T & 0 & 0 & 0 & 0 \\ \frac{\partial \boldsymbol{\Phi}}{\partial \boldsymbol{\gamma}} \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{v}} & 0 & \frac{\partial \boldsymbol{\Phi}}{\partial \boldsymbol{\kappa}} & 0 & 0 \\ \frac{\partial \boldsymbol{\tau}}{\partial \mathbf{v}} & 0 & 0 & \frac{\partial \boldsymbol{\tau}_p}{\partial \boldsymbol{\eta}_p} - \frac{\partial \boldsymbol{\tau}_p}{\partial \boldsymbol{\alpha}} & -\frac{\partial \boldsymbol{\tau}_p}{\partial \boldsymbol{\alpha}} \\ 0 & 0 & 0 & -\frac{\partial \boldsymbol{\tau}_p}{\partial \boldsymbol{\eta}_p} & \mathbf{I} \end{pmatrix}$$

Some computations

Recall that

$$f_a = \int_E s_a q n de \quad s_a(e) = \begin{cases} 1 & \text{if } a = e_1 \\ -1 & \text{if } a = e_2 \\ 0 & \text{otherwise} \end{cases}$$

Let $w = x_{e_2} - x_{e_1}$ the vector connecting the endpoints of a truss. Then

$$\frac{\partial f_a}{\partial v_b} = -\Delta t \int_E s_a s_b \frac{\partial(qn)}{\partial w} de.$$

Lemma

Let $w = x_b - x_a$ then

$$\frac{\partial \tau}{\partial w} = \frac{\partial \tau}{\partial \epsilon_e} \frac{n}{l}$$

and

$$\frac{\partial(qn)}{\partial w} = \frac{V}{l^2} \left(\left(\frac{\partial \tau}{\partial \epsilon_e} - 2\tau \right) n \otimes n + \tau I_{d \times d} \right).$$

Proof.

$$\frac{\partial l}{\partial w} = n \quad \text{and} \quad \frac{\partial \epsilon}{\partial w} = \frac{1}{l} \frac{\partial l}{\partial w} = \frac{n}{l} \quad \text{and} \quad \frac{\partial n}{\partial w} = \frac{I_{d \times d} - n \otimes n}{l}$$

and

$$\frac{\partial(qn)}{\partial w} = -\frac{V}{l^2} \tau n \otimes n + \frac{V}{l^2} \frac{\partial \tau}{\partial \epsilon_e} n \otimes n + \frac{V}{l^2} \tau (I_{d \times d} - n \otimes n)$$

□

Examples

Bodies and deformations

let $\alpha \in \{S, M\}$ the deformation is

$$\chi^\alpha: \Omega^\alpha \rightarrow \mathbb{R}^d$$

we decompose

$$\partial\Omega^\alpha = \Gamma_D^\alpha \cup \Gamma_N^\alpha \cup \Gamma_C^\alpha$$

Dirichlet condition

on Γ_D^α we impose

$$\chi^\alpha = \bar{\chi}^\alpha$$

let $r^\alpha: \Gamma_D \rightarrow \mathbb{R}^d$ the
corresponding Lagrange multiplier

Projection

defined as $\Pi: \Gamma_C^S \times [0, T] \rightarrow \Gamma_C^M$ such that

$$\Pi(X^S, t) \in \operatorname{argmin}_{X^M \in \Gamma_C^M} |\chi^S(X_S, t) - \chi^M(X_M, t)|$$

Gap vector and gap distance

the gap vector is

$$g(X^S, t) = \chi^S(X^S, t) - \chi^M(\Pi(X^S, t), t)$$

and the gap distance is

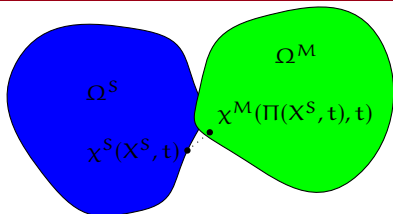
$$\gamma(X^S, t) = g(X^S, t) \cdot n^M(\chi^M(\Pi(X^S, t), t), t)$$

notice that $g = \gamma n^M$

Contact condition

let $\kappa: \Gamma_C^S \rightarrow \mathbb{R}$ the Lagrange multiplier

$$\gamma \geq 0 \quad \kappa \geq 0 \quad \gamma \kappa = 0$$



Lemma

Let $v^\alpha = \dot{\chi}^\alpha$ and $F^\alpha = \text{Grad } \chi^\alpha$. Then

$$\dot{g} = v^S - v^M - F^M \dot{\Gamma}.$$

Moreover let $K^M = -\text{grad}_{\gamma_C^M} n^M$ be the curvature tensor of γ_C^M . Then

$$g = \gamma n \quad \dot{g} = n^M \dot{\gamma} - K^M F^M \dot{\Gamma} + \dot{n}^M$$

and

$$\dot{\gamma} = (v^S - v^M) \cdot n^M$$

Proof.

Take the material time derivative and use that the derivative of a unit vector is orthogonal to the unit vector itself □

space of deformation

$$\mathcal{X}^\alpha = W^{1,p}(\Omega^\alpha)^d$$

space of virtual velocities

$$\mathcal{V}^\alpha = W^{1,p}(\Omega^\alpha)^d$$

space of Dirichlet reactions

$$\mathcal{R}^\alpha = \left[W^{1-\frac{1}{p},p}(\Gamma_D^\alpha)^d \right]^*$$

cone of admissible gap distances

$$\mathcal{G} = W^{1-\frac{1}{p},p}(\Gamma_C^S) \quad \mathcal{C} = \{\gamma \in \mathcal{G} | \gamma \geq 0\}$$

cone of admissible contact reactions

$$\mathcal{K} = \mathcal{G}^* \quad \mathcal{D} = \{\kappa \in \mathcal{K} | \langle \kappa, \tilde{\gamma} \rangle \geq 0 \text{ for all } \tilde{\gamma} \in \mathcal{C}\}$$

Weak form of the contact condition

Lemma

Let $\gamma \in \mathcal{C}$. Then $\kappa \in N_{\mathcal{C}}(\gamma)$ if and only if $\kappa = 0$ in $\Gamma_I^S(\gamma)$ and $\kappa \geq 0$ in $\Gamma_A^S(\gamma)$.

Proof.

We have $\kappa \in N_{\mathcal{C}}(\gamma)$ if and only if

$$\int_{\Gamma_C^S} \kappa(\tilde{\gamma} - \gamma) dA = \int_{\Gamma_A^S(\gamma)} \kappa \tilde{\gamma} dA + \int_{\Gamma_I^S(\gamma)} \kappa(\tilde{\gamma} - \gamma) dA \geq 0$$

for all $\tilde{\gamma} \in \mathcal{C}$. Take $\tilde{\gamma} \geq 0$ arbitrary with support in $\Gamma_A^S(\gamma)$ so $\kappa \geq 0$ there. On $\Gamma_I^S(\gamma)$ we have $\gamma > 0$.

Take $\tilde{\gamma} \geq 0$ arbitrary with support in $\Gamma_I^S(\gamma)$ so $\kappa = 0$ there. □

Weak form of the contact condition

Lemma

The condition $\gamma \in \mathcal{C}$ and $\kappa \in N_{\mathcal{C}}(\gamma)$ is equivalent to $\kappa \in \mathcal{D}$ and $\gamma \in N_{\mathcal{D}}(\kappa)$.

Proof.

Using a theorem from Convex Analysis the condition $\gamma \in \mathcal{C}$ and $\kappa \in N_{\mathcal{C}}(\gamma)$ is equivalent to $-\kappa \in \text{dom } \sigma_{\mathcal{C}}$ and $\gamma \in \partial^s \sigma_{\mathcal{C}}(-\kappa)$. Here $\sigma_{\mathcal{C}} = \iota_{\mathcal{C}}^*$ is the support function of $\mathcal{C}$. We have

$$\sigma_{\mathcal{C}}(-\kappa) = \sup_{\tilde{\gamma} \in \mathcal{C}} (-\langle \kappa, \tilde{\gamma} \rangle).$$

Since $\tilde{\gamma} \in \mathcal{C}$ implies $t\tilde{\gamma} \in \mathcal{C}$ for all $t \geq 0$ then $-\kappa \in \text{dom } \sigma_{\mathcal{C}}$ if and only if $\kappa \in \mathcal{D}$. Moreover $\sigma_{\mathcal{C}}$ is the indicator function of $-\mathcal{D}$. So $\gamma \in \partial^s \sigma_{\mathcal{C}}(-\kappa)$ if and only if $\gamma \in N_{\mathcal{D}}(\kappa)$. □

Remark

Formally $\gamma \in N_{\mathcal{D}}(\kappa)$ means

$$\int_{\Gamma_C^S} (\tilde{\kappa} - \kappa) \gamma \, dA \geq 0$$

for all $\tilde{\kappa} \in \mathcal{D}$

Internal virtual power

$$\tilde{W} = \sum_{\alpha} \int_{\Omega^{\alpha}} \mathbf{P}^{\alpha} : \text{Grad } \tilde{\mathbf{v}}^{\alpha} dX - \int_{\Gamma_C^S} \kappa \tilde{\gamma} dA.$$

can be rewritten as

$$\begin{aligned} \tilde{W} = & \sum_{\alpha} \left[- \int_{\Omega^{\alpha}} \text{div}(\mathbf{P}^{\alpha}) \cdot \tilde{\mathbf{v}}^{\alpha} dX + \int_{\Gamma_D^{\alpha}} \mathbf{P}^{\alpha} \mathbf{N}^{\alpha} \cdot \tilde{\mathbf{v}}^{\alpha} dA + \int_{\Gamma_N^{\alpha}} \mathbf{P}^{\alpha} \mathbf{N}^{\alpha} \cdot \tilde{\mathbf{v}}^{\alpha} dA + \right. \\ & + \int_{\Gamma_I^{\alpha}} \mathbf{P}^{\alpha} \mathbf{N}^{\alpha} \cdot \tilde{\mathbf{v}}^{\alpha} dA \Big] + \\ & + \int_{\Gamma_A^S} \left(\mathbf{P}^S \mathbf{N}^S + \kappa \mathbf{n}^S(\chi^S, t) \right) \cdot \tilde{\mathbf{v}}^S dA + \\ & + \int_{\Gamma_A^M} \left(\mathbf{P}^M \mathbf{N}^M + C_M^S \kappa(\Pi^{-1}, t) \mathbf{n}^M(\chi^M, t) \right) \cdot \tilde{\mathbf{v}}^M dA \end{aligned}$$

External virtual power and balances

$$\tilde{\mathcal{P}} = \sum_{\alpha} \left[\int_{\Omega^{\alpha}} f^{\alpha} \cdot \tilde{v}^{\alpha} dX + \int_{\Gamma_D^{\alpha}} r^{\alpha} \cdot \tilde{v}^{\alpha} dA + \int_{\Gamma_N^{\alpha}} h^{\alpha} \cdot \tilde{v}^{\alpha} dA \right].$$

we get the balances

$$\operatorname{div}(\mathbf{P}^{\alpha}) + f^{\alpha} = 0 \quad \text{on } \Omega^{\alpha}$$

$$\mathbf{P}^{\alpha} \mathbf{N}^{\alpha} = \mathbf{r}^{\alpha} \quad \text{on } \Gamma_D^{\alpha}$$

$$\mathbf{P}^{\alpha} \mathbf{N}^{\alpha} = \mathbf{h}^{\alpha} \quad \text{on } \Gamma_N^{\alpha}$$

$$\mathbf{P}^{\alpha} \mathbf{N}^{\alpha} = 0 \quad \text{on } \Gamma_I^{\alpha}$$

$$\mathbf{N}^{\alpha} = -\kappa \mathbf{n}^S(\chi^S, t) \quad \text{on } \Gamma_A^S$$

$$\mathbf{P}^M \mathbf{N}^M = -C_M^S \kappa(\Pi^{-1}, t) \mathbf{n}^M(\chi^M, t) \quad \text{on } \Gamma_A^M$$

Initial boundary value problem

Problem

Let $\bar{\chi}^\alpha(X^\alpha, t)$ with $\bar{\chi}^\alpha(X^\alpha, 0) = X^\alpha$, $f^\alpha(X^\alpha, t)$ and $h^\alpha(X^\alpha, t)$ be given. For all $t \in [0, T]$ find $\chi^\alpha(t) \in \mathcal{X}^\alpha$ with $\chi^\alpha(X^\alpha, 0) = X^\alpha$, $r^\alpha(t) \in \mathcal{R}^\alpha$ and $\kappa(t) \in \mathcal{D}$ such that

$$\tilde{W} = \tilde{P} \quad \text{for all } \tilde{v}^\alpha \in \mathcal{V}^\alpha$$

$$\int_{\Gamma_D^\alpha} (\chi^\alpha - \bar{\chi}^\alpha) \cdot \tilde{r}^\alpha dA = 0 \quad \text{for all } \tilde{r}^\alpha \in \mathcal{R}^\alpha$$

$$\int_{\Gamma_C^S} (\tilde{\kappa} - \kappa) \gamma dA \geq 0 \quad \text{for all } \tilde{\kappa} \in \mathcal{D}.$$

Numerical approximation

We take the deformation of the form

$$\chi(X) = N(X)\chi \quad \chi \in \mathbb{R}^N \quad N: \Omega \rightarrow \mathbb{R}^{d \times N}$$

and the virtual velocity

$$\tilde{v}(X) = N(X)\tilde{v} \quad \tilde{v} \in \mathbb{R}^N$$

Moreover

$$r(X) = L(X)r \quad \kappa(X) = M(X)\kappa$$

with

$$r \in \mathbb{R}^L \quad \kappa \in \mathbb{R}^M$$

and

$$L: \Gamma_D \rightarrow \mathbb{R}^{d \times L} \quad M: \Gamma_C \rightarrow \mathbb{R}^{1 \times M}$$

We redefine the function spaces accordingly and we set

$$\mathcal{D} = \{ \kappa \in \mathcal{K} \mid \kappa_\alpha \geq 0 \text{ for } \alpha \in \{1, \dots, M\} \}.$$

$$\int_{\Omega} \hat{\mathbf{P}} \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \boldsymbol{\chi} \right) : \frac{\partial \mathbf{N}}{\partial \mathbf{X}} \tilde{\mathbf{v}} d\mathbf{X} - \int_{\Gamma_C} \mathbf{M} \boldsymbol{\kappa} \mathbf{G} \tilde{\mathbf{v}} d\mathbf{A} = \int_{\Omega} \mathbf{f} \cdot \mathbf{N} \tilde{\mathbf{v}} d\mathbf{X} + \int_{\Gamma_D} \mathbf{L} \mathbf{r} \cdot \mathbf{N} \tilde{\mathbf{v}} d\mathbf{A} + \int_{\Gamma_N} \mathbf{h} \cdot \mathbf{N} \tilde{\mathbf{v}} d\mathbf{A}$$

gives

$$\begin{aligned} \mathbf{R}_1 = \int_{\Omega} \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \right)^T : \hat{\mathbf{P}} \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \boldsymbol{\chi} \right) d\mathbf{X} - \int_{\Gamma_C} \mathbf{G}^T \mathbf{M} \boldsymbol{\kappa} d\mathbf{A} - \\ - \int_{\Omega} \mathbf{N}^T \mathbf{f} d\mathbf{X} - \int_{\Gamma_D} \mathbf{N}^T \mathbf{L} \mathbf{r} d\mathbf{A} - \int_{\Gamma_N} \mathbf{N}^T \mathbf{h} d\mathbf{A} = 0. \end{aligned}$$

$$\int_{\Gamma_D} (\mathbf{N}\chi - \bar{\chi}) \cdot \mathbf{L}\tilde{\mathbf{r}} dA = 0$$

gives

$$\mathbf{R}_2 = \int_{\Gamma_D} \mathbf{L}^T (\mathbf{N}\chi - \bar{\chi}) dA = 0$$

$$0 \leq \int_{\Gamma_C} (\tilde{\kappa} - \kappa) \gamma \, dA = \gamma^T (\tilde{\kappa} - \kappa) \quad \gamma = \int_{\Gamma_C^S} M^T \gamma \, dA$$

is equivalent to

$$\gamma_a \geq 0 \quad \kappa_a \geq 0 \quad \gamma_a \kappa_a = 0$$

which we write as

$$\mathbf{R}_3 = \Phi(\gamma, \kappa) = 0$$

Residual and stiffness matrix

The residual and stiffness matrix have the form

$$\mathbf{R} = \begin{pmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \mathbf{R}_3 \end{pmatrix} \quad \mathbf{K} = \begin{pmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} & \mathbf{K}_{13} \\ \mathbf{K}_{21} & 0 & 0 \\ \mathbf{K}_{31} & 0 & \mathbf{K}_{33} \end{pmatrix}.$$

We have that

$$\mathbf{K}_{11} = \Delta t \int_{\Omega} \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \right)^T : \hat{\mathbf{P}} \frac{\partial \mathbf{N}}{\partial \mathbf{X}} d\mathbf{X} + \Delta t \int_{\Omega} \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \right)^T : \left(\frac{\partial \hat{\mathbf{P}}}{\partial \mathbf{F}} \frac{\partial \mathbf{N}}{\partial \mathbf{X}} \right) \left(\frac{\partial \mathbf{N}}{\partial \mathbf{X}} \mathbf{x} \right) d\mathbf{X}$$

$$\mathbf{K}_{12} = - \int_{\Gamma_D} \mathbf{N}^T \mathbf{L} dA$$

$$\mathbf{K}_{13} = - \int_{\Gamma_C} \mathbf{G}^T \mathbf{M} dA.$$

Moreover

$$\mathbf{K}_{21} = \int_{\Gamma_D} \mathbf{L}^T \mathbf{N} dA$$

and

$$\mathbf{K}_{31} = \frac{\partial \Phi}{\partial \gamma} \frac{\partial \gamma}{\partial \mathbf{v}}, \quad \mathbf{K}_{33} = \frac{\partial \Phi}{\partial \kappa}.$$

We notice that

$$\frac{\partial \gamma}{\partial \mathbf{v}} = \Delta t \frac{\partial \Phi}{\partial \gamma} \int_{\Gamma_C} \mathbf{M}^T \mathbf{G} dA.$$

Example

Example